Cantera vs neural networks | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

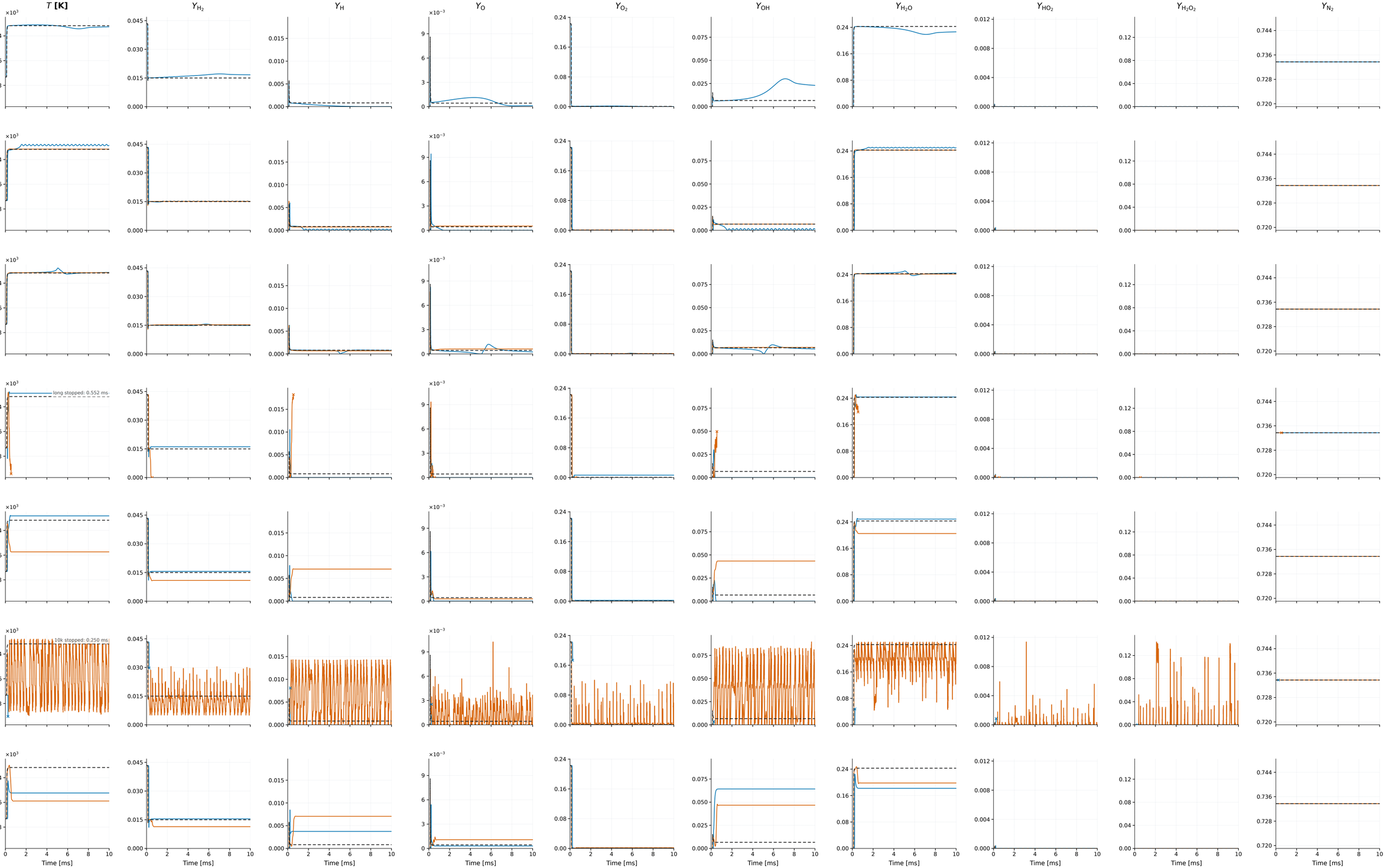
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

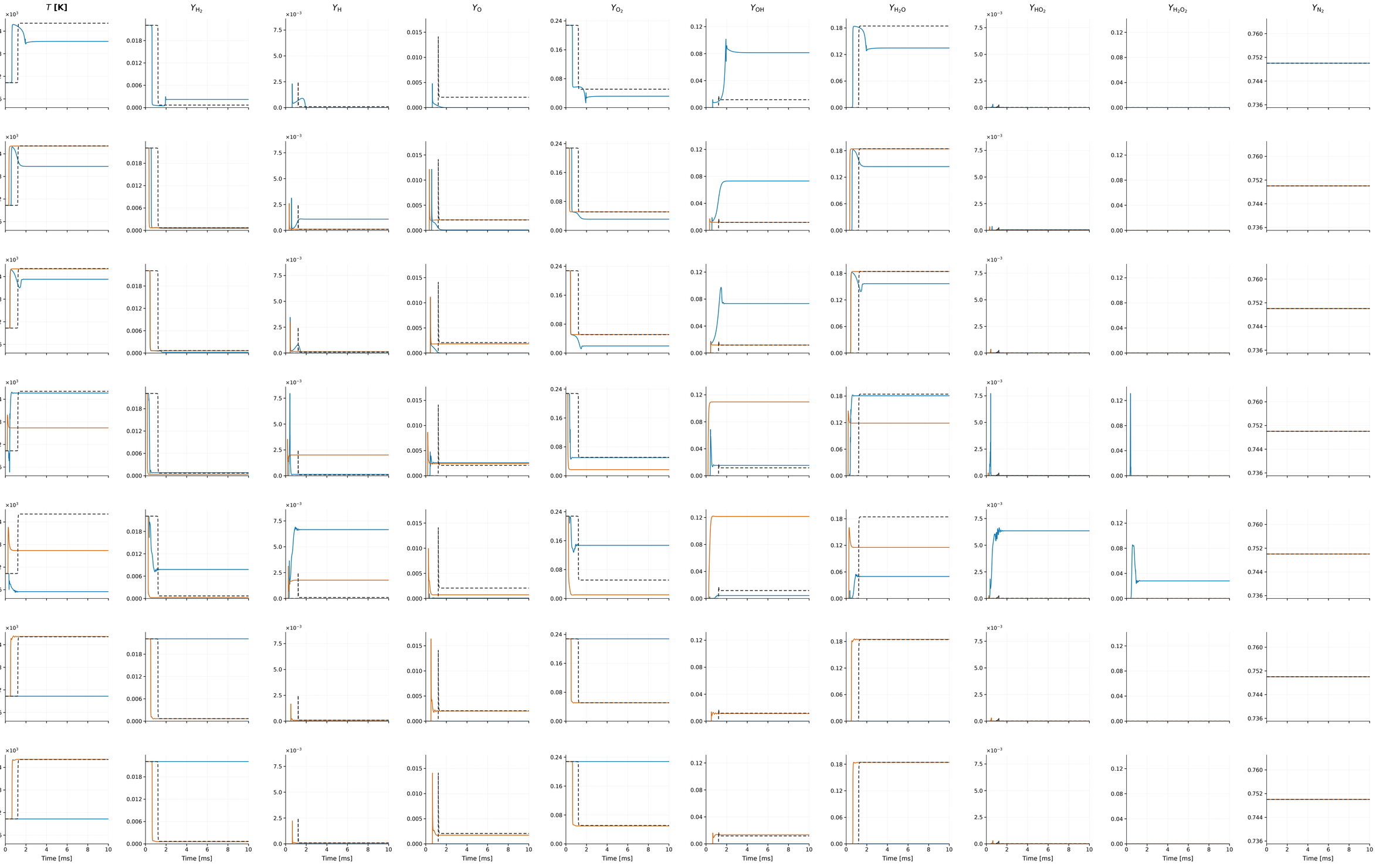
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

long stopped: 0.486 ms

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10
10k updates

2 P-only
DeepONet10

3 ANN10 +
P-only DeepONet10

4 T/P
DeepONet10

5 ANN10 +
T/P DeepONet10

6 T/P/ϕ
DeepONet10

7 ANN10 +
T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

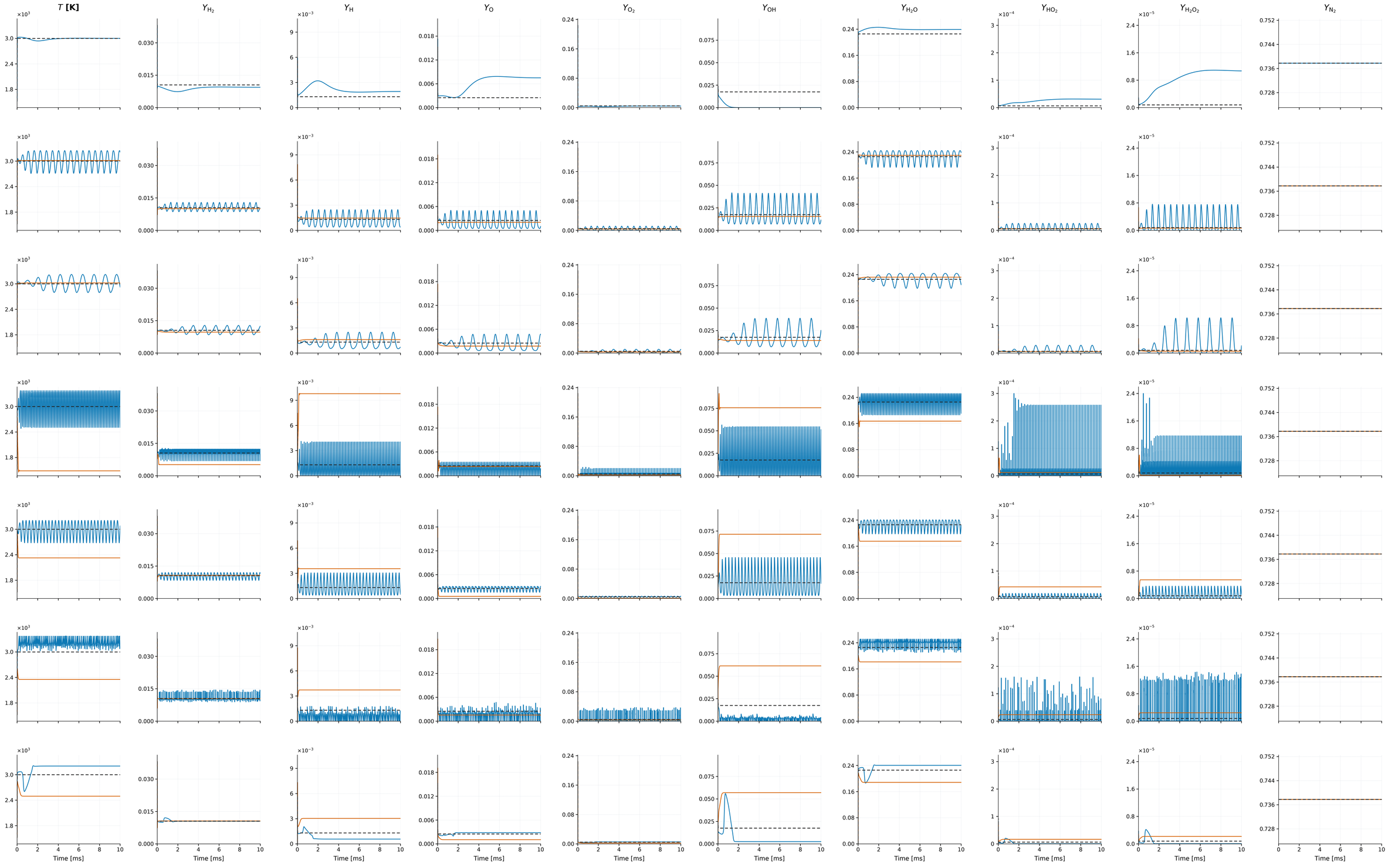
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10
10k updates

2 P-only
DeepONet10

3 ANN10 +
P-only DeepONet10

4 T/P
DeepONet10

10k stopped: 1.084 ms

5 ANN10 +
T/P DeepONet10

6 T/P/ϕ
DeepONet10

7 ANN10 +
T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs neural networks | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.